E-PROM — User Guide

Express Fuzzy Preference Ranking with PROMETHEE

E-PROM combines three building blocks:

```
FAHP-Express  →  fuzzy criterion weights  →  Fuzzy PROMETHEE  →  ranking
                                                    |
                                  Monte Carlo (CPP-style) stability analysis
```

Each decision maker (DM) evaluates the alternatives and supplies a short set of importance judgments. E-PROM runs the whole method **once per DM**, keeps every quantity fuzzy, aggregates the DMs' flows, and defuzzifies only at the very end.

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1. Quick start

1. Fill the **evaluation workbook** (one sheet per DM) and the **FAHP-Express workbook** (one sheet per DM).
2. Upload both files in the sidebar.
3. For each criterion choose: data type, direction, preference function, and `q` / `p`.
4. Click **Run E-PROM** → ranking, flows, PROMETHEE I relations.
5. Open the **Monte Carlo** tab to test how stable the ranking is.
6. Download the Excel / CSV results.

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2. Input files

2.1 Evaluation workbook

Sheet	Content
One sheet per DM (sheet name = DM name)	Column A: alternative ID · Column B: description · Column C onward: criteria
Sheets named `Guide`, `_Lists` or `Instructions`	Support sheets, ignored by the app. Every other sheet is read as a decision maker.

Rules:

- All DMs must use the same alternative IDs and the same criterion names. The *order* of rows and criteria may differ between sheets (they are aligned by ID and by name; names ignore case and accents).
- Ordinal criteria: integers inside the scale you select (**5, 7 or 9** — a 1–10 scale is not supported).
- Continuous criteria: any real number, in the original unit (cost in R$, schedule or time in months, distance, consumption, physical performance, ...). Values are never converted to a 1–7 scale.
- No empty cells.

ID	Description	Impact (ordinal 1–9)	Cost (continuous, R$)	Deadline (continuous, months)
R1	Alternative 1	7	300000	12
R2	Alternative 2	5	250000	18

2.2 FAHP-Express workbook

One sheet per DM, plus the ignored `Guide` / `_Lists` sheets. Each DM sheet has:

Row label	What to fill
`Criteria`	The criterion names (same names as the evaluation workbook). **The first criterion is that DM's reference criterion.**
`Comparison with reference`	For every criterion, how many times more important the reference is than that criterion (linguistic term, see §5). The reference itself is *Equal importance*.
`Attribute precision`	`High`, `Medium` or `Low` for every criterion (see §4). A blank cell is treated as `High`.

The sheets must have the same names as in the evaluation workbook: decision makers are paired **by sheet name** (if the names differ, they are paired by position and the app shows a warning). The criteria must be the same as in the evaluation workbook. Comparison values can be linguistic terms (English or Portuguese) or the numbers 1–9; precision labels also accept the Portuguese forms `Alta`, `Média`, `Baixa`. The numeric fuzzy parameters of each precision level are fixed internally — you only choose the label.

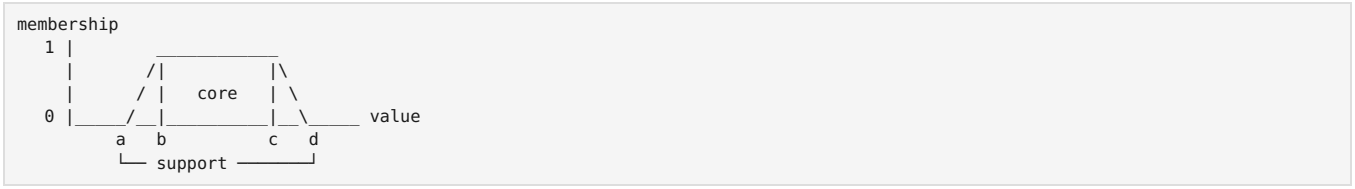
Each DM may choose a **different reference criterion**. The reference should be the criterion that DM considers most important, so every other factor is ≥ 1.

> `Attribute precision` is stored in the FAHP-Express sheet for convenience, but it describes the **evaluations** of that DM on that criterion. It does **not** change the weights.

3. Fuzzy numbers — what they are and why E-PROM uses them

3.1 Trapezoidal fuzzy number

A judgment such as "about 7" is not a point. E-PROM represents it as a trapezoid `[a, b, c, d]`:



- **Core** `[b, c]` — values that are fully compatible with the judgment (membership 1).
- **Support** `[a, d]` — values that are possible at all (membership > 0).
- A crisp number is the special case `a = b = c = d`.
- The wider the trapezoid, the more uncertain the judgment.

3.2 Arithmetic used

For $A = [a_1, b_1, c_1, d_1]$ and $B = [a_2, b_2, c_2, d_2]$:

Operation	Rule
Addition	$[a_1+a_2,\ b_1+b_2,\ c_1+c_2,\ d_1+d_2]$
Subtraction (interval arithmetic)	$[a_1-d_2,\ b_1-c_2,\ c_1-b_2,\ d_1-a_2]$
Division by a scalar s	$[a_1/s,\ b_1/s,\ c_1/s,\ d_1/s]$
Multiplication (weight × preference)	Dubois–Prade / Geldermann LR approximation: modal values multiply (b_1b_2, c_1c_2); spreads follow $left = b_1\alpha_2 + b_2\alpha_1 - \alpha_1\alpha_2$, $right = c_1\beta_2 + c_2\beta_1 + \beta_1\beta_2$, where $\alpha = b-a$ and $\beta = d-c$ (negative spreads are set to 0)
Defuzzification (Center of Area)	$\bar{x} = [(c+d)^2 - cd - (a+b)^2 + ab] \,/\, [3(c+d-a-b)]$

Subtraction widens the result (the width of $A - B$ is the sum of both widths). This is intentional: the difference between two uncertain values is itself uncertain.

3.3 Where fuzziness enters E-PROM

Stage	Fuzzy object	Defuzzified?
Evaluations (ordinal and continuous)	trapezoid built from the observed value and its precision (§4)	No
Criterion weights	trapezoid from FAHP-Express (§5)	No
Pairwise preference	`weight ⊗ preference` (§6)	No
Flows Φ^+, Φ^-, Φ (per DM)	trapezoids	No
Group flows (mean over DMs)	trapezoids	**Yes — once, by COA**, to rank

4. Precision — how uncertain is each evaluation?

Precision answers: "*how sure is this DM about this evaluation on this criterion?*" It is set **per DM × criterion** (`Attribute precision` row) and uses three labels:

Label	Meaning	Fuzzy width
High	The DM is confident / the data is exact	narrow
Medium	Default linguistic representation	intermediate
Low	The DM is unsure / the data is rough	wide

Precision is monotone: **High ⇒ narrower trapezoid ⇒ less uncertainty.** The three labels act on different scales depending on the criterion type.

4.1 Ordinal criteria (scales 5, 7, 9) — widths in scale units

Each ordinal value v has a base trapezoid (the **Medium** representation):

9-point scale

v	[a, b, c, d]	v	[a, b, c, d]
1	[1, 1, 1.5, 2]	6	[5, 5.5, 6.5, 7]
2	[1, 1.5, 2.5, 3]	7	[6, 6.5, 7.5, 8]
3	[2, 2.5, 3.5, 4]	8	[7, 7.5, 8.5, 9]
4	[3, 3.5, 4.5, 5]	9	[8, 8.5, 9, 9]
5	[4, 4.5, 5.5, 6]		

7-point scale: 1 = [1, 1, 1.5, 2] · 2 = [1, 1.5, 2.5, 3] · 3 = [2, 2.5, 3.5, 4] · 4 = [3, 3.5, 4.5, 5] · 5 = [4, 4.5, 5.5, 6] · 6 = [5, 5.5, 6.5, 7] · 7 = [6, 6.5, 7, 7]

5-point scale: 1 = [1, 1, 1.5, 2] · 2 = [1.5, 2, 2, 2.5] · 3 = [2, 2.5, 3, 3.5] · 4 = [3, 3.5, 4, 4.5] · 5 = [4, 4.5, 5, 5]

Precision then **contracts or dilates the base trapezoid around** `v`:

```
fuzzy = v + factor × (base − v),   then clipped to [1, scale max]
factor:  High = 0.5   Medium = 1.0   Low = 1.5
```

Example — `v = 7` on the 9-point scale:

Precision	Trapezoid
High	[6.5, 6.75, 7.25, 7.5]
Medium	[6, 6.5, 7.5, 8]
Low	[5.5, 6.25, 7.75, 8.5]

Values at the ends of the scale are truncated by the clipping, so their trapezoids are one-sided. **The same number means different things on different scales** (a 5 on a 5-point scale is the top; a 5 on a 9-point scale is the middle) — always select the scale the DMs actually used.

4.2 Continuous criteria — relative width in %

The value stays in its **original unit**. A value `x` becomes

```
[ x(1 − r_s),  x(1 − r_c),  x(1 + r_c),  x(1 + r_s) ]
 r_s = support half-width (outer)     r_c = core half-width (inner)
```

Precision	Support r_s	Core r_c
High	±5 %	±2 %
Medium	±10 %	±4 %
Low	±16.67 %	±6.67 %

Example — a cost of R$ 30,000:

Precision	Trapezoid (R$)
High	[28,500, 29,400, 30,600, 31,500]
Medium	[27,000, 28,800, 31,200, 33,000]
Low	[25,000, 28,000, 32,000, 35,000]

Notes:

- The width is **proportional to the value** — expensive alternatives carry a wider absolute band than cheap ones.
- A value of exactly `0` has no width (it stays `[0, 0, 0, 0]`).
- Use **High** for contractual/measured data, **Medium** for engineering estimates, **Low** for rough or expert-guess figures.

4.3 How precision interacts with `q` and `p`

The preference function is applied to the **fuzzy difference** of two alternatives, not to the difference of the crisp values. For two alternatives with the same precision on a continuous criterion, the fuzzy difference spans roughly `± 2·r_s·x` around the crisp difference.

Practical consequences:

- If `q` is **smaller** than that spread (≈ `2·r_s·x`), two nearly equal alternatives will still get a small, non-zero preference in *both* directions. It cancels out in the net flow Φ, but it inflates Φ^+ and Φ^- (and can affect PROMETHEE I relations).
- Rule of thumb: choose `q` **at least as large as** `2·r_s·x` for typical values `x` if you want near-ties to be true indifference.
- With the **Usual** function on ordinal criteria, identical ratings produce a preference of about 0.5 in each direction for the same reason. This is expected behavior of fuzzy PROMETHEE.

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5. FAHP-Express — fuzzy criterion weights

5.1 The "Express" idea

Classical AHP asks $n(n-1)/2$ pairwise comparisons. FAHP-Express asks **only $n-1$ per DM**: every criterion is compared with a single reference criterion.

For a DM with values $v_1 \dots v_n$ (reference has $v = 1$):

```
A_ij = v_j / v_i        (A_1j = v_j,   A_j1 = 1 / v_j)
```

So the complete comparison matrix is rebuilt from one row. It is **perfectly consistent by construction** (consistency ratio = 0), so there is no consistency check to perform.

v_j reads: "*the reference criterion is v_j times more important than criterion j.*"

5.2 Linguistic scale and its fuzzy form

v	Linguistic term	Trapezoid
1	Equal importance	[1, 1, 1, 1]
2	Weakly more important	[1, 1.5, 2.5, 3]
3	Moderately more important	[2, 2.5, 3.5, 4]
4	Moderately to strongly more important	[3, 3.5, 4.5, 5]
5	Strongly more important	[4, 4.5, 5.5, 6]
6	Strongly to very strongly more important	[5, 5.5, 6.5, 7]
7	Very strongly more important	[6, 6.5, 7.5, 8]
8	Very strongly to extremely more important	[7, 7.5, 8.5, 9]
9	Extremely more important	[8, 8.5, 9, 9]

- Ratios $A_{ij} = v_j / v_i$ are generally **not integers** (e.g. 6/4 = 1.5). They are fuzzified by **linear interpolation** between the two neighbouring rows of the table.
- Ratios below 1 use the reciprocal: $[1/d, 1/c, 1/b, 1/a]$.
- Any ratio outside $[1/9, 9]$ is rejected.

5.3 Weights (Buckley)

For each criterion i:

1. r_i = geometric mean of row i of the fuzzy matrix (computed component-wise).
2. $w_i = r_i \otimes (\Sigma\ r)^{-1}$, where $(\Sigma\ r)^{-1} = [1/\Sigma d, 1/\Sigma c, 1/\Sigma b, 1/\Sigma a]$.

The result is a **fuzzy weight** (trapezoid) per criterion and per DM. A crisp weight (mean of the four numbers, normalized to 1) is computed only for display and for the Monte Carlo weight summary. **Fuzzy weights are what actually enter PROMETHEE.**

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6. Fuzzy PROMETHEE

6.1 Criterion configuration

Type	Allowed functions	Notes
Ordinal	Usual · Level	q, p in *scale units*
Continuous	U-shape · V-shape · Linear	q, p in the *original unit*

Direction: *Maximize* (more is better) or *Minimize* (less is better, e.g. cost).

6.2 Preference functions (for a difference $d \geq 0$)

Function	$P(d)$	Parameters
Usual	0 if $d \leq 0$, else 1	none
U-shape	0 if $d \leq q$, else 1	$q \geq 0$
V-shape	d/p if $d < p$, else 1	$p > 0$
Level	0 if $d \leq q$; 0.5 if $q < d \leq p$; 1 if $d > p$	$0 \leq q < p$
Linear	0 if $d \leq q$; $(d-q)/(p-q)$ if $q < d < p$; 1 if $d \geq p$	$0 \leq q < p$

- q — **indifference threshold**: differences up to q are ignored.
- p — **preference threshold**: differences from p on are a strict preference.

- Always express `q` and `p` in the criterion's original unit. Example: cost in R$: `q = 10000`, `p = 50000`; deadline in months: `q = 1`, `p = 4`.

6.3 Calculation (per DM)

For every ordered pair of alternatives `(i, j)`:

1. **Fuzzy difference** on criterion `k`: `D = F_ik ⊖ F_jk` (for *Minimize*: `F_jk ⊖ F_ik`).
2. **Fuzzy preference**: the preference function is applied to the four corners of `D` (the function is non-decreasing, so the result is again a trapezoid `P_k`).
3. **Weighted preference**: `π(i,j) = Σₖ  w_k ⊗ P_k`.
4. **Flows**:
   - `Φ⁺(i) = (1/(n−1)) Σⱼ π(i,j)` — how much `i` outranks the others;
   - `Φ⁻(i) = (1/(n−1)) Σⱼ π(j,i)` — how much `i` is outranked;
   - `Φ(i) = Φ⁺(i) ⊖ Φ⁻(i)` — net flow.

6.4 Group aggregation

The DMs' fuzzy flows are **averaged in the fuzzy domain** (a-posteriori aggregation); the group `Φ⁺`, `Φ⁻` and `Φ` are then defuzzified by COA. The ranking sorts alternatives by the defuzzified net flow Φ (rank 1 = highest Φ; ties share the average rank).

6.5 PROMETHEE I relations

Using the defuzzified `Φ⁺` and `Φ⁻`:

Relation	Condition
P (i outranks j)	`Φ⁺(i) ≥ Φ⁺(j)` and `Φ⁻(i) ≤ Φ⁻(j)`, with at least one strict
I (indifferent)	both flows equal
R (incomparable)	`i` is better on one flow and worse on the other

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7. Monte Carlo (CPP-style) stability analysis

A single deterministic ranking hides how much it depends on uncertain inputs. The Monte Carlo module repeats E-PROM many times, each time drawing the uncertain inputs from probability distributions, and reports **probabilities** instead of a single answer. It is a *probabilistic preference* analysis in the spirit of CPP (Composition of Probabilistic Preferences): the input is a distribution, the output is the probability of each ranking outcome.

7.1 What is simulated in each iteration

Input	How it is drawn
Ordinal evaluation (per alternative × criterion × DM)	**Empirical distribution** — one of the answers given by the DMs for that alternative and criterion, drawn with its observed frequency. Example: answers `7, 7, 6, 5, 7, 6, 7, 4, 7, 6` → P(7)=0.5, P(6)=0.3, P(5)=0.1, P(4)=0.1.
Continuous evaluation	**Uniform** in `[x(1−v), x(1+v)]`, where `v` is the % variation you set per criterion (e.g. 20 % → `[0.8x, 1.2x]`).
Criterion weights	Each iteration **bootstraps the decision makers**: it draws, with replacement, as many DMs as there are in the panel, takes their complete FAHP-Express rows, and averages their fuzzy weights. Whole rows are resampled, so it stays consistent when DMs use different reference criteria. The same weight vector is used for all DMs in that iteration. With very few DMs (e.g. 2) there are only a few distinct weight combinations, so the weight variability is limited. A legacy mode (each criterion draws its own value) is available; use it only if every DM has the same reference criterion.
Precision	Unchanged; each DM keeps their own precision for the fuzzification.

Then, in each iteration: fuzzy PROMETHEE per DM → average of the fuzzy flows → COA → ranking.

> **Two kinds of uncertainty.** *Precision* (§4) is the DM's imprecision about a judgment and lives *inside* the fuzzy number. The *Monte Carlo variation* is a scenario analysis *around* it. For continuous criteria they act together, so a large variation on top of Low precision is very conservative.

7.2 Outputs

Output	How to read it
Mean position / SD of position	Average rank over the simulations and its dispersion. Small SD = stable position.
First-place frequency	% of simulations in which the alternative is ranked 1st (ties for 1st share the credit).
Mean Φ / SD of Φ	Average net flow and its dispersion.
Rank acceptability matrix	% of simulations in which alternative *i* ends in position *r*. Each row sums to 100 %. Alternatives tied in a simulation share the tied positions equally.
Outranking acceptability (PROMETHEE I)	% of simulations in which *i* outranks *j* (P relation).
Weight distribution	Mean, SD, min and max of each criterion's crisp weight across simulations.

7.3 Number of simulations

The Monte Carlo error of a probability `p` is about `√(p(1−p)/N)`. With `N = 2000` it is ≤ 1.1 percentage points. Use ≥ 1000 for exploration and ≥ 5000 for results you will report. The seed makes runs reproducible.

7.4 How to interpret

- Use the analysis as a **stability check**, not as a replacement for the deterministic ranking.
- A deterministic leader with a high first-place frequency (and a narrow rank distribution) is a **robust** choice.
- Two alternatives with similar first-place frequencies and a low outranking acceptability in both directions are **not distinguishable** under the stated uncertainty — report them as a tie group.
- If the weight distribution is very wide for a criterion, the DMs disagree about its importance; consider a consensus round.

E-PROM is domain-generic: it has no `KEY` classification and no military-domain groups (those belong to SIPREM). It produces a ranking and PROMETHEE I relations.

Templates: `E_PROM_Evaluations_Template.xlsx` and `E_PROM_FAHP_Express_Template.xlsx` (downloadable inside the app). English and Portuguese labels are both accepted in the workbooks.

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8. Recommended workflow

1. Define alternatives and criteria; decide for each criterion its type (ordinal / continuous) and direction.
2. Collect each DM's evaluations (same IDs, same criterion names).
3. Collect each DM's FAHP-Express row (reference + comparisons) and the **attribute precision**.
4. Upload both workbooks and configure the criteria (function, `q`, `p`, scale).
5. Check `q` and `p` against the fuzzy widths of §4.3.
6. Run the deterministic E-PROM; inspect ranking, Φ per DM and PROMETHEE I relations.
7. Compare the DMs' individual Φ — large disagreement is a signal to discuss before aggregating.
8. Run the Monte Carlo; inspect first-place frequency, rank acceptability and weight spread.
9. Report the deterministic ranking **together with** its stability.
10. Export the results.

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9. Glossary

Term	Meaning
TFN / trapezoid	Fuzzy number `[a, b, c, d]` with support `[a, d]` and core `[b, c]`.
COA	Center of Area — defuzzification that returns the x-coordinate of the trapezoid's centroid.
Φ^+, Φ^-, Φ	Positive (leaving), negative (entering) and net PROMETHEE flows.
Reference criterion	The criterion each DM compares all others against in FAHP-Express (first column).
Attribute precision	High / Medium / Low confidence of a DM in the evaluations of one criterion.
q / p	Indifference / preference thresholds of the preference function.
Rank acceptability	Probability that an alternative occupies a given rank.
CPP	Composition of Probabilistic Preferences: probabilistic treatment of preferences when inputs are distributions.